

第六章 不定积分

§6.1 原函数 & 不定积分

定义 (原函数)

在区间 I 上给函数 $f(x)$, 若存在定义在 I 上的函数 $F(x)$ 使 $F'(x) = f(x) \quad \forall x \in I$ 则称 $F(x)$ 为 $f(x)$ 的一个原函数

• $F(x) + C$ 也为原函数

定义 (不定积分)

若 $f(x)$ 在区间 I 上存在原函数, 则称 $f(x)$ 的所有原函数为 $f(x)$ 的不定积分记为

$$\int f(x) dx = F(x) + C$$

Peano 余项中, 不要漏 $(x \rightarrow x_0) !!!$

$\int$ 为积分符号. $f(x)$ 为被积函数. x 为积分变量.

积分表. 必 C - 一定不要漏!!!

$$\int x^\alpha dx = \frac{1}{\alpha+1} x^{\alpha+1} + C \quad (\alpha \neq -1) \quad \alpha = -1: \int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

指数函数

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \frac{dx}{\cos^2 x} = \tan x + C$$

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$\int \sinh x = \cosh x + C$$

$$\int \cosh x = \sinh x + C$$

$$\int \frac{dx}{\cosh^2 x} = \tanh x + C$$

做题方法: ① 归类.

② 换元. 变成我熟悉的函数形式

③ 往积分表上凑.

用于处理式

带根号的不定积分 (注意)

换元

$\arcsin x$ 处理

引理. 设 $f(x), g(x)$ 都 $\exists$ 原函数, 且 $k \in \mathbb{R}$ 为常数. 则运算

• (I) $f(x) \pm g(x)$ 也存在原函数. 且 $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$

• (II) $k f(x)$ 也存在原函数. 且 $\int k f(x) dx = k \int f(x) dx$

例. 求 $\int \frac{x^4}{1+x^2} dx$ 通分.

$$\int \frac{x^4}{1+x^2} dx = \int \left(x^2 + \frac{1}{x^2+1} \right) dx = \int (x^2) dx + \int \frac{1}{x^2+1} dx = \frac{1}{3} x^3 - x + \arctan x + C$$

例. 求 $\int \tan^2 x dx$ 典型

$$\int \tan^2 x dx = \int \left(\frac{1}{\cos^2 x} - 1 \right) dx = \tan x - x + C$$

$\sec x = \frac{1}{\cos x}$

§ 6.2 换元法和分部积分法.

定理 (第一换元法)

设 $f(u)$ 的原函数为 $F(u) + C$ 且 $u = u(x)$ 关于 x 可导.

则有 $\int f(u(x)) u'(x) dx = F(u(x)) + C$

例. 求 $\int \frac{1}{x-1} dx$.
 $\frac{1}{2} u(x) = \frac{1}{x-1}$
 $f(u) = \frac{1}{u} \quad \frac{1}{x-1} = f(u(x)) u'(x)$

令 $t(x) = x-1 \quad \therefore \int \frac{1}{t(x)} t'(x) dx = \ln|t(x)| + C = \ln|x-1| + C$
 $\frac{dt}{dx} = 1 \quad \therefore dt = dx$

例. 求 $\int (ax+b)^n dx \quad (a \neq 0, n \neq -1)$
 $\frac{1}{2} u(x) = ax+b \quad u'(x) = a \quad \text{即} \quad \frac{du}{dx} = a \Rightarrow dx = \frac{du}{a}$
 $(ax+b)^n = f(u(x)) \cdot u'(x)$ 作换元代换

$\therefore \int (ax+b)^n dx = \int u^n \cdot \frac{1}{a} du = \frac{1}{a} \int u^n du = \frac{1}{a(n+1)} u^{n+1} + C = \frac{(ax+b)^{n+1}}{a(n+1)} + C$

例. 求 $\int \frac{1}{a^2+x^2} dx \quad (a \neq 0)$

$\frac{1}{2} u(x) = x+ia^2 \quad \frac{du}{dx} = 2x \quad \therefore dx = \frac{du}{2x}$

$\int \frac{1}{a^2+x^2} dx = \int \frac{1}{u(x)} \frac{du}{2x} = \frac{1}{2} \ln|u(x)| + C = \frac{1}{2} \ln(x^2+a^2) + C$ 换元要作出代换.

$\int \frac{1}{a^2+x^2} dx = \frac{1}{a^2} \int \frac{1}{1+(\frac{x}{a})^2} dx$ 令 $u = \frac{x}{a} \quad \frac{du}{dx} = \frac{1}{a} \Rightarrow du = \frac{dx}{a}$
 $\frac{1}{a^2} \int \frac{1}{1+u^2} du = \frac{1}{a^2} \arctan u + C = \frac{1}{a} \arctan \frac{x}{a} + C$

$\therefore \int \frac{1}{a^2+x^2} dx = \frac{1}{a^2} \int \frac{1}{1+u^2} du \cdot a = \frac{1}{a} \arctan u + C = \frac{1}{a} \arctan \frac{x}{a} + C$
 得到 $\frac{1}{1+x^2}$ 积分.

例. 求 $\int \frac{x}{1+x^2} dx$.

令 $u = x^2 \quad \frac{du}{dx} = 2x \quad \therefore dx = \frac{du}{2x} \quad \therefore \int \frac{x}{1+x^2} dx = \int \frac{x}{1+u} \cdot \frac{du}{2x} = \frac{1}{2} \int \frac{1}{1+u} du = \frac{1}{2} \ln|1+u| + C = \frac{1}{2} \ln(1+x^2) + C$

例. 求 $I_n = \int \tan^n x dx \quad (n \in \mathbb{Z}^+)$
 ① 变形 $\frac{\sin x}{\cos x}$ ② 用 $\tan x$ 换元.

法一. 令 $\tan x = u(x) \quad \frac{du}{dx} = \frac{1}{\cos^2 x}$ 即 $du = \frac{1}{\cos^2 x} dx = (u^2+1) dx$

$\therefore I_n = \int u^n \frac{du}{u^2+1} = \int \frac{u^n}{u^2+1} du$ ③

法二. 当 $n=1$ 令 $u = \cos x \quad du = -\sin x \cdot dx \quad \therefore \int \frac{\sin x}{\cos x} dx = \int \frac{\sin x}{u} \cdot \frac{du}{-\sin x} = -\int \frac{1}{u} du = -\ln|u| + C = -\ln|\cos x| + C$

对于 $n \geq 2$. 记 $I_n(x) = \int \tan^n x dx$

有 $\int \tan^n x dx = \int \tan^{n-2} x \tan^2 x dx$ 提出 $\tan^2 x + 1 = \frac{1}{\cos^2 x} = (\tan x)'$ 提出 $\tan^2 x$.

$= \int \tan^{n-2} x (1 + \tan^2 x) dx - \int \tan^{n-2} x dx = \frac{1}{n-1} \tan^{n-1}(x) - I_{n-2}(x) + C$

$\hookrightarrow \frac{1}{\cos^2 x} = (\tan x)' = \frac{d \tan x}{dx} \quad \therefore (1 + \tan^2 x) dx = d \tan x$

$\therefore \int \tan^{n-2} x (1 + \tan^2 x) dx = \int \tan^{n-2} x d \tan x = \frac{1}{n-1} \tan^{n-1} + C.$

可以递归得到 $I_n(x)$

$I_n = \int \tan^n x dx$ ④ 变形 $\frac{\sin x}{\cos x}$

令 $u(x) = \tan x$ ⑤ $\frac{du}{dx} = \frac{1}{\cos^2 x} \quad \therefore dx = du \cdot \cos^2 x, \cos^2 x = \frac{1}{1+u^2} = \frac{1}{1+u^2(x)}$

$\int = u^n(x) \cdot du \cdot \cos^2 x = \int u^n(x) \frac{du}{1+u^2(x)} \quad t(x) = 1+u^2(x) \quad \frac{dt}{du} = 2u \Rightarrow du = \frac{dt}{2u}$

$\int u^n(x) \frac{dt}{2u} = \frac{1}{2} \int \frac{t^{n/2}}{t} dt$

例. $\int \frac{dx}{x^4-1}$
 看到高次, 要化简成低次.

$$\int \frac{dx}{x^4-1} = \frac{1}{2} \int \left[\frac{1}{x^2-1} - \frac{1}{x^2+1} \right] dx \quad \text{其中 } \int \frac{dx}{x^2-1} = \frac{1}{2} \int \left[\frac{1}{x-1} - \frac{1}{x+1} \right] dx$$

$$= \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$$

例. $\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \int \left[\frac{1}{a-x} + \frac{1}{a+x} \right] dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$

例. 求 $I = \int \frac{1}{1+x^2} dx$ $J = \int \frac{x}{1+x^2} dx$ **★但看到都有高次. (除非有明显暗示, 换元可解决) 首先要因式分解.**

$$I+J = \int \frac{x+1}{x^2+1} dx = \int \frac{1}{x^2+1} dx = \int \frac{dx}{(x-\frac{1}{2})^2 + \frac{3}{4}} = \frac{4}{3} \int \frac{dy}{(x-\frac{1}{2})^2 + 1}$$

$$\text{取 } u = \frac{2\sqrt{3}}{3}(x-\frac{1}{2}) \quad \frac{du}{dx} = \frac{2\sqrt{3}}{3} \quad \therefore \text{原式} = \frac{4}{3} \int \frac{\frac{\sqrt{3}}{2} du}{u^2+1} = \frac{2\sqrt{3}}{3} \arctan \frac{2x-1}{\sqrt{3}} + C$$

★看到不可因式分解的二次式, 就配成 $(\quad)^2 + \text{常数}$

$$I-J = \int \frac{1-x+x^2}{x^3+1} dx = \int \frac{x^2}{x^3+1} dx = \int \frac{1}{1+u} \frac{du}{3} = \frac{1}{3} \ln|1+u| = \frac{1}{3} \ln|x^3+1| + C$$

例. (1) $\int \frac{dx}{\sqrt{a^2-x^2}}$ (2) $\int \frac{dx}{\sqrt{x(1-x)}}$
 看到根号. ① 第=种换元 ② ① $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$ 上靠.

(1) $\int \frac{1}{|a|} \frac{dx}{\sqrt{1-(\frac{x}{a})^2}} \quad \frac{1}{2} u(x) = \frac{x}{a}. \quad \frac{du}{dx} = \frac{1}{a} \quad \therefore dx = a du. \quad \frac{1}{|a|} \int \frac{dx}{\sqrt{1-(\frac{x}{a})^2}} = \frac{1}{|a|} \int \frac{a \cdot du}{\sqrt{1-u^2}} = \frac{a}{|a|} \arcsin \frac{x}{a} + C$

(2) $\int \frac{dx}{\sqrt{x(1-x)}} = \int \frac{dx}{\sqrt{\frac{1}{4} - (x-\frac{1}{2})^2}} = 2 \int \frac{dx}{\sqrt{1-4(x-\frac{1}{2})^2}} \quad \frac{1}{2} u(x) = 2x-1 \quad \therefore \frac{du}{dx} = 2.$

$$x \in (0,1) \quad 2 \int \frac{\frac{du}{2}}{\sqrt{1-u^2}} = \arcsin(2x-1) + C.$$

只差一个常数!!!

(如果要证明, 只需证两导数相同即可)

源: 利用 $\sqrt{x}$. 令 $t = \sqrt{x}$ 则 $\frac{dt}{dx} = \frac{1}{2\sqrt{x}} = \frac{1}{2t} \Rightarrow dx = 2t dt$

$$\text{原式} = \int \frac{2t \cdot dt}{t \cdot \sqrt{1-t^2}} = 2 \int \frac{dt}{\sqrt{1-t^2}} = 2 \arcsin t + C$$

嘉淑原. 因为 $\frac{dt}{dx} = \frac{1}{2\sqrt{x}} = \frac{1}{2t}$ 导数出来, 可用新变量 $t = \sqrt{x}$ 表示.

例. $\int \frac{1}{\cos^3 x} dx$

$\frac{1}{2} y = \sin x$. 则 $dy = \cos x \cdot dx$. $\therefore \text{原式} = \int \frac{1}{\cos^3 x} \cdot \cos x \cdot dx = \int \frac{1}{1-y^2} dy = \frac{1}{2} \ln \left| \frac{1+y}{1-y} \right| + C$

$$= \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C.$$

换成 $\cos x$ 不好办. 上面会多出一个 $\sin x$.

嘉淑原. 对偶式. $\therefore \cos x$ 变成 $\frac{dx}{\cos^2 x - \sin^2 x} = 2 \int \frac{\frac{d(\frac{x}{2})}{\cos^2 \frac{x}{2}}}{(1-\tan^2 \frac{x}{2})} = 2 \int \frac{d \tan \frac{x}{2}}{1-\tan^2 \frac{x}{2}} = \ln \left| \frac{1+\tan \frac{x}{2}}{1-\tan \frac{x}{2}} \right| = \ln \left| \tan^2 \frac{x}{2} + \frac{3}{4} \right| + C$

$$\int \frac{dt}{1-t^2} = \frac{1}{2} \int \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt.$$

$$= \frac{1}{2} (\ln|1+t| - \ln|1-t|) = \frac{1}{2} \ln \left| \frac{1+t}{1-t} \right| + C.$$

例. $I = \int x^3 \ln x dx$ 微分不易 积分易 $x^3 \cdot \ln x, \arctan x, \arcsin x$ 等.
 $u(x) = \ln x \quad v'(x) = x^3 \Rightarrow v(x) = \frac{1}{4} x^4$ 喜欢导数形式. 不知道积分.

$$\therefore I = \int u(x) v'(x) dx = u(x) v(x) - \int u'(x) v(x) dx = \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + C$$

例. $I = \int \arctan x dx$

$$u(x) = \arctan x \quad v'(x) = 1 \Rightarrow v(x) = x$$

$$I = \int u(x) v'(x) dx = uv - \int v(x) u'(x) dx = x \arctan x - \int \frac{x}{x^2+1} dx$$

$$\text{取 } x^2+1=t \quad \text{则 } \frac{dt}{dx} = 2x \Rightarrow dx = \frac{dt}{2x} \quad \therefore \int \frac{x}{x^2+1} dx = \int \frac{x}{t} \cdot \frac{dt}{2x} = \frac{1}{2} \int \frac{dt}{t} = \frac{1}{2} \ln|t| = \frac{1}{2} \ln(x^2+1) + C$$

$$\therefore I = x \arctan x - \frac{1}{2} \ln(x^2+1) + C$$

例. $I = \int x^2 \sin x dx$ 希望降低幂次 $\rightarrow$ 通过让幂函数一直作本分部积分中求导的那一部分

$$u(x) = x^2 \quad v'(x) = \sin x \Rightarrow v(x) = -\cos x$$

$$\therefore I = \int u(x) v'(x) dx = u(x) v(x) - \int u'(x) v(x) dx = -x^2 \cos x + \int 2x \cos x dx$$

$$\text{求 } \int x \cos x. \quad \frac{1}{2} u(x) = x \quad v'(x) = \cos x \Rightarrow v(x) = \sin x \quad \text{再次分部积分. } \rightarrow u(x) = x, v'(x) = \cos x$$

$$\therefore \int x \cos x = x \sin x - \int \sin x dx = x \sin x + \cos x + C$$

$$\therefore I = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

例. $\int x^2 e^x dx$ $\frac{dx^2}{dx} = 2x$

$$I = \int x^2 de^x = x^2 e^x - \int e^x dx^2 = x^2 e^x - \int 2x \cdot e^x dx$$

$$\int 2x de^x = 2x \cdot e^x - \int e^x d2x = 2x e^x - 2 \int e^x dx = 2x e^x - 2e^x + C$$

$$\therefore I = x^2 e^x - 2x e^x + 2e^x + C$$

例. $I = \int \sqrt{x^2-1} dx$

$$\text{法一. } \frac{1}{2} v'(x) = 1 \quad u(x) = \sqrt{x^2-1} \quad \therefore I = x \sqrt{x^2-1} - \int v(x) u'(x) dx = x \sqrt{x^2-1} - \int \frac{x^2}{\sqrt{x^2-1}} dx$$

$$= x \sqrt{x^2-1} - \int \frac{1}{\sqrt{x^2-1}} dx - \int \sqrt{x^2-1} dx = x \sqrt{x^2-1} - \int \frac{1}{\sqrt{x^2-1}} dx - I.$$

$$\therefore I = \frac{1}{2} (x \sqrt{x^2-1} - \int \frac{1}{\sqrt{x^2-1}} dx) \quad \text{由前面, } \int \frac{1}{\sqrt{x^2-1}} dx = \ln|x + \sqrt{x^2-1}| + C$$

$$\therefore I = \frac{1}{2} (x \sqrt{x^2-1} - \ln|x + \sqrt{x^2-1}|) + C$$

法二: $\frac{1}{2} x = \sec t \in (0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2}) \Rightarrow t$ 取这样的范围. 保证可以讨论到 $x > 1$ & $x < -1$ 两种可能
 同时保证 $\tan t > 0$. 下只讨论 $x > 1$. 即 $t \in (0, \frac{\pi}{2})$ 情况

$$\text{则 } \frac{dx}{dt} = (\frac{1}{\cos t})' = \frac{\sin t}{\cos^2 t}$$

$$\therefore I = \int \sqrt{x^2-1} dx = \int \tan t \frac{\sin t}{\cos^2 t} dt = \int \frac{\sin^2 t}{\cos^3 t} dt = \int \frac{\sin^2 t}{(1-\sin^2 t)^{\frac{3}{2}}} \cos t dt \quad \frac{1}{2} s = \sin t$$

$$\text{有 } I = \int \frac{s^2}{(1-s^2)^2} ds = \int \frac{1}{s^2-1} ds + \int \frac{1}{(s^2-1)^2} ds \quad \frac{s^2}{(1-s^2)^2} = \frac{s^2+1}{(1-s^2)^2} = \frac{1}{s^2-1} + \frac{1}{(1-s^2)^2}$$

$$\Rightarrow \int \frac{1}{s^2-1} ds = \ln|\frac{s-1}{s+1}| + C \quad \int \frac{1}{(s^2-1)^2} ds = \int [\frac{1}{2} (\frac{1}{s-1} - \frac{1}{s+1})]^2 ds = \frac{1}{4} \int [\frac{1}{(s-1)^2} + \frac{1}{(s+1)^2} - \frac{1}{s-1} + \frac{1}{s+1}] ds$$

$$= \frac{1}{4} [-(s-1)^{-1} - (s+1)^{-1} - \ln|s-1| + \ln|s+1|] + C$$

未定成

例. $I = \int e^{ax} \cos bx dx$ $J = \int e^{ax} \sin bx dx$

1. $u(x) = \cos bx$ $v'(x) = e^{ax} \Rightarrow v(x) = \frac{1}{a} e^{ax}$

$I = \frac{1}{a} e^{ax} \cos bx - \int \frac{1}{a} e^{ax} (-b \sin bx) dx = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int e^{ax} \sin bx dx = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} J + C \dots (1)$

2. $u(x) = \sin bx$ $v'(x) = e^{ax} \Rightarrow v(x) = \frac{1}{a} e^{ax}$

$J = \frac{1}{a} e^{ax} \sin bx - \int \frac{1}{a} e^{ax} b \cos bx dx = \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos bx dx = \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} I + C \dots (2)$

联立 (1) (2) 解得 $I = \frac{e^{ax} (a \cos bx + b \sin bx)}{a^2 + b^2} + C$ $J = \frac{e^{ax} (a \sin bx - b \cos bx)}{a^2 + b^2} + C$

证法二: $I = \frac{1}{2} \int e^{ax} (e^{ibx} + e^{-ibx}) dx = \frac{1}{2} \left(\frac{e^{(ib+a)x}}{a+ib} + \frac{e^{(a-ib)x}}{a-ib} \right) + C$
 $= \frac{1}{2} e^{ax} \frac{(a-ib)(\cos bx + i \sin bx) + (a+ib)(\cos bx - i \sin bx)}{a^2 + b^2} + C = \frac{1}{2} e^{ax} \frac{2a \cos bx + 2b \sin bx}{a^2 + b^2} + C$

§ 6.3 其他类型函数的不定积分

有理函数的积分

有理函数指两个多项式之比. $R(x) = \frac{P(x)}{Q(x)}$ 其中 $P(x), Q(x)$ 为互质多项式

真分式: 分子次数小于分母 假分式: 分子次数大于分母. 所有假分式可写成真分式 + 多项式形式

$Q(x) = (x-a)^k \dots (x-b)^t (x^2+px+q)^l \dots (x^2+rx+s)^n$ 其中 $p^2-4q < 0 \dots r^2-4s < 0$

(n 次方程只存在实数解和共轭复数解)

$\therefore$ 真分式 $\frac{P(x)}{Q(x)}$ 可分解为 $\frac{P(x)}{Q(x)} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_k}{(x-a)^k} + \dots + \frac{B_1}{x-b} + \frac{B_2}{(x-b)^2} + \dots + \frac{B_t}{(x-b)^t}$
 $+ \frac{C_1x+D_1}{x^2+px+q} + \dots + \frac{C_lx+D_l}{(x^2+px+q)^l} + \dots + \frac{E_1x+F_1}{x^2+rx+s} + \dots + \frac{E_nx+F_n}{(x^2+rx+s)^n}$

其中各项称为最简真分式

证: 若 $m(x)$ 和 $n(x)$ 互素.

例 $\exists u(x), v(x)$ st $u(x)m(x) + v(x)n(x) = 1 \therefore \frac{P(x)}{Q(x)} = \frac{P(x)}{m(x)n(x)} = P(x) \left(\frac{u(x)}{m(x)} + \frac{v(x)}{n(x)} \right)$

$\therefore$ 最简真分式仅有以下四种形式: ① $\frac{A}{x-a}$ ② $\frac{A}{(x-a)^m}$ ③ $\frac{Ax+B}{x^2+px+q}$ ④ $\frac{Ax+B}{(x^2+px+q)^m}$

最简真分式的积分

① $\int \frac{A}{x-a} = A \ln|x-a| + C$ ② $\int \frac{A}{(x-a)^m} = \frac{A}{1-m} \frac{1}{(x-a)^{m-1}} + C$

③ $\int \frac{Ax+B}{x^2+px+q} = \int \frac{A(x+\frac{p}{2}) + B - \frac{1}{2}Ap}{(x+\frac{p}{2})^2 + q - \frac{1}{4}p^2} = \int \frac{A(x+\frac{p}{2})}{(x+\frac{p}{2})^2 + q - \frac{1}{4}p^2} + \int \frac{B - \frac{1}{2}Ap}{(x+\frac{p}{2})^2 + q - \frac{1}{4}p^2}$

对 $I = \int \frac{mx}{x^2+n} dx$ 令 $t = x^2$ $\frac{dt}{dx} = 2x$ 则 $I = \int \frac{mx}{t+n} \frac{dt}{2x} = \frac{m}{2} \int \frac{dt}{t+n} = \frac{m}{2} \ln|t+n| = \frac{m}{2} \ln|x^2+n|$

对 $J = \int \frac{m}{x^2+n} dx = \frac{m}{n} \int \frac{1}{\frac{x^2}{n} + 1} dx$ 令 $t = \frac{x}{\sqrt{n}}$ 则 $\frac{dt}{dx} = \frac{1}{\sqrt{n}}$ $\therefore J = \frac{m}{n} \int \frac{1}{t^2+1} \sqrt{n} dt = \frac{m}{\sqrt{n}} \arctan t = \frac{m}{\sqrt{n}} \arctan \frac{x}{\sqrt{n}}$

④ $\int \frac{(Ax+B) dx}{(x^2+px+q)^m} = \int \frac{Ax}{(x+\frac{p}{2})^2 + q - \frac{p^2}{4}} dx + \int \frac{B}{(x+\frac{p}{2})^2 + q - \frac{p^2}{4}} dx$ 归结为上面例题形式

例. $\star I_n = \int \frac{dx}{(x^2+a^2)^n} \rightarrow$ 这种有几次方的, 不建这个事件. $\frac{1}{11}$ 的思路?

令 $v(x)=1 \Rightarrow v(x)=x \quad u(x)=\frac{1}{(x^2+a^2)^n}$

$$\therefore I_n = \frac{x}{(x^2+a^2)^n} - \int v(x) u'(x) dx = \frac{x}{(x^2+a^2)^n} - \int x \cdot (-n) \cdot 2x \cdot (x^2+a^2)^{-n-1} dx = \frac{x}{(x^2+a^2)^n} + 2n \int \frac{x^2}{(x^2+a^2)^{n+1}} dx$$

$$= \frac{x}{(x^2+a^2)^n} + 2n I_n - 2n a^2 I_{n+1}$$

$\therefore$ 递推公式 $I_{n+1} = \frac{1}{2na^2} \frac{x}{(x^2+a^2)^n} + \frac{n-1}{2a^2} I_n$ 特别. $I_2 = \int \frac{dx}{(x^2+a^2)^2} = \frac{x}{2a^2(x^2+a^2)} + \frac{1}{2a^2} \int \frac{dx}{x^2+a^2}$

$$= \frac{x}{2a(x^2+a^2)} + \frac{1}{2a^3} \arctan \frac{x}{a} + C = \frac{1}{a} \arctan \frac{x}{a}$$

例. $\int \frac{x}{(x^2+1)(x-1)} dx = -\frac{1}{2} \int \left(\frac{x-1}{x^2+1} - \frac{1}{x-1} \right) dx$

$$\int \frac{x-1}{x^2+1} dx = \frac{1}{2} \int \frac{2x \cdot dx}{x^2+1} - \int \frac{1}{x^2+1} dx = \frac{1}{2} \int \frac{d(x^2+1)}{x^2+1} - \int \frac{1}{x^2+1} dx = \frac{1}{2} \ln(x^2+1) - \arctan x + C$$

$$\int \frac{1}{x-1} dx = \ln|x-1| + C$$

$$\therefore I = -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \arctan x + \frac{1}{2} \ln|x-1| + C$$

例 $\int \frac{x^2+1}{x^4+x^3+x^2-x} dx = \int \frac{x^2+1}{x(x-1)^3} dx$

令 $\frac{x^2+1}{x(x-1)^3} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{D}{(x-1)^3} = \frac{(A+B)x^3 + (C-3A-2B)x^2 + (B+3A-C)x + D-A}{x(x-1)^3}$

代原式满足.

$$\begin{cases} A+B=1 \\ C-3A-2B=0 \\ B+3A-C=0 \\ D-A=1 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=0 \\ C=3 \\ D=2 \end{cases}$$

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三角有理函数积分.

三角有理函数: $P(u, v)$ 和 $Q(u, v)$ 均为 二元多项式, 三角函数有理函数为形如.

$$R(\sin x, \cos x) = \frac{P(\sin x, \cos x)}{Q(\sin x, \cos x)}$$

通过换元法, 可证三角有理函数积分可以化为有理函数积分.

万能变换. 令 $t = \tan \frac{x}{2}$ 则 $x = 2 \arctan t \Rightarrow dx = \frac{2}{t^2+1} dt$

万能公式 $\begin{cases} \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} = 2 \tan \frac{x}{2} \cos^2 \frac{x}{2} = \frac{2t}{1+t^2} \\ \cos x = 2 \cos^2 \frac{x}{2} - 1 = \frac{2}{t^2+1} - 1 = \frac{1-t^2}{1+t^2} \end{cases}$

$$\therefore I = \int R(\sin x, \cos x) dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2} dt. \text{ 化为有理函数积分的公式}$$

• 会有比万能公式更简单的换元法。

1. $R(u, -u) = -R(u, u)$ 必有 $R(u, u) = u R_1(u^2)$ 成立。令 $t = \sin x$

有 $\int R_1(\sin x, \cos x) \cos x dx = \int R_1(t, 1-t^2) dt$ 得到关于 t 的有理函数积分。

2. $R(-u, u) = -R(u, u)$ 必有 $R(u, u) = u R_1(u^2)$ 成立。令 $t = \cos x$

有 $\int R_1(\sin^2 x, \cos x) \sin x dx = - \int R_1(1-t^2, t) dt$

3. $R(-u, -u) = R(u, u)$ 必有 $R(u, u) = R_1(\frac{u}{u^2}, u^2)$ 成立。令 $t = \tan x$

有 $\int R_1(\tan x, \sec^2 x) dx = - \int R_1(t, \frac{1}{1+t^2}) \frac{1}{1+t^2} dt$

例. $\int \frac{dx}{5+4\sin x}$

$$\begin{aligned} \text{令 } t = \tan \frac{x}{2} \text{ 则 } dx &= \frac{2}{t^2+1} dt \quad \therefore I = \int \frac{\frac{2}{t^2+1} dt}{5 + \frac{8t}{t^2+1}} = 2 \int \frac{dt}{5t^2 + 8t + 5} \\ &= 2 \int \frac{dt}{(\sqrt{5}t + \frac{4}{\sqrt{5}})^2 + \frac{9}{5}} = \frac{9}{10} \int \frac{dt}{\frac{5}{9}(\sqrt{5}t + \frac{4}{\sqrt{5}})^2 + 1} = \frac{2}{3} \arctan(\frac{5}{3}t + \frac{4}{3}) + C \end{aligned}$$

例. $\frac{1}{2} \int \frac{1-r^2}{1-2\cos x + r^2} dx \quad (0 < r < 1)$ 几何意义: $(1-r\cos\theta)^2 + (0-r\sin\theta)^2$
三角函数和常数在一起, 大概能用万能公式

令 $t = \tan \frac{x}{2} \Rightarrow dx = \frac{2}{1+t^2} dt$

$$I = \int \frac{1-r^2}{(1-r)^2 + (1+r)^2 t^2} dt = \int \frac{dt \cdot \frac{1-r}{1+r}}{\frac{1-r}{1+r} + \frac{1+r}{1-r} t^2} = \frac{1}{1+r} \int \frac{dt}{1 + (\frac{1+r}{1-r} t)^2} = \int \frac{\frac{1+r}{1-r} dt}{1 + (\frac{1+r}{1-r} t)^2} = \arctan(\frac{1+r}{1-r} t)$$

例. (1) $I = \int \frac{\cos^3 x}{1+\sin x} dx$ (2) $I = \frac{\sin x \cos x}{\sin^2 x + \cos x} dx$

(1) 令 $t = \sin x$ 则 $\frac{dt}{dx} = \cos x \quad \therefore I = \int \frac{1-t^2}{1+t} dt = \int (-1 + \frac{2}{1+t}) dt = -t + 2 \arctan t + C$

(2) 令 $t = \cos x$ 则 $\frac{dt}{dx} = -\sin x \quad \therefore I = \int \frac{\sin x \cdot t}{(1-t^2) + t} \frac{dt}{-\sin x} = \int \frac{-t}{t^2 - t - 1} dt$

法一: $t^2 - t - 1$ 因式分解, 令出根 $\frac{1 \pm \sqrt{5}}{2}$.

法二: $\frac{1}{2} t^2 - t - 1 = m \quad \therefore \frac{dm}{dt} = 2t - 1 \quad \therefore I = \frac{1}{2} \int \frac{(2t-1)dt}{t^2-t-1} + \frac{1}{2} \int \frac{dt}{t^2-t-1}$
 $= \frac{1}{2} \int \frac{d(t^2-t-1)}{t^2-t-1} + \frac{1}{2\sqrt{5}} \int (\frac{1}{t - \frac{1-\sqrt{5}}{2}} - \frac{1}{t + \frac{1-\sqrt{5}}{2}})$

$\bar{p} = \bar{q}$:

① 分部积分。

② 把又右又讨厌。

而东西给换元

① 分部积分, 出现自己
 ② 用 $\arcsin x$.
 ③ 把 "sqrt" 大换元

例. $\int \frac{\cos^2 x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2} dx = \frac{1}{2} \ln |t^2 - t - 1| + \frac{\sqrt{5}}{10} \ln \left| \frac{t - \frac{1-\sqrt{5}}{2}}{t + \frac{1-\sqrt{5}}{2}} \right| + C$

$\therefore R(-u, -u) = R(u, u) \quad \therefore \frac{1}{2} t = \tan x. \quad \frac{dt}{dx} = \frac{1}{\cos^2 x}$

$\therefore I = \int \frac{\cos^4 x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2 \cos^2 x} dx = \int \frac{dt}{(a^2 t^2 + b^2)^2}$ 拆上面提过的分母积分。

无理函数积分 I

- 考虑如下类型的积分: $I = \int R(x, \sqrt[m]{\frac{ax+b}{cx+d}}) dx$ 其中 R 为二元有理函数. 且 $ad-bc \neq 0$
- 令 $t = \sqrt[m]{\frac{ax+b}{cx+d}}$ 则有 $x = \frac{dt^m - b}{a - ct^m}$ $dx = \frac{m(ad-bc)t^{m-1}}{(a-ct^m)^2} dt$

于是有 $I = \int R\left(\frac{dt^m - b}{a - ct^m}, t\right) \frac{m(ad-bc)t^{m-1}}{(a-ct^m)^2} dt$ 化为有理函数积分

例. $I = \int \frac{1 - \sqrt{x+1}}{1 + \sqrt[3]{x+1}} dx$

令 $t = (x+1)^{\frac{1}{6}}$ 则 $x = t^6 - 1$ $dx = 6t^5 dt$

则 $I = \int \frac{1-t^3}{1+t^2} 6t^5 dt = 6 \int \frac{t^5 - t^8}{1+t^2} dt = 6 \int (1-t-t^2+t^3+t^4-t^6 + \frac{t-1}{t^2+1} dt)$
 $= 6t - 3t^2 - 2t^3 + \frac{3}{2}t^4 + \frac{6}{5}t^5 - \frac{6}{7}t^7 + 3\ln(t^2+1) - 6\arctan t$

无理函数积分 II

- 考虑如下类型的二项式微分式积分 $I = \int x^m (a+bx^n)^p dx$, a, b 常, m, n, p 有理
- 先令 $x^n = t$. $dx = \frac{1}{n} t^{\frac{1}{n}-1} dt$. $\therefore I = \frac{1}{n} \int t^{\frac{m+1}{n}-1} (a+bt)^p dt$ 记 $\frac{m+1}{n}-1 = q$.
- 则求 $\int t^q (a+bt)^p dt$
- 化成以下三种情况 (除这三种之外, 原函数不为初等函数).

1. p 为整数. 记 $q = \frac{j}{k}$, $j, k \in \mathbb{Z}^+$. 令 $u = t^{\frac{1}{k}}$, 则可化为有理函数的积分

2. q 为整数. 记 $p = \frac{j}{k}$, $j, k \in \mathbb{Z}^+$. 令 $u = (a+bt)^{\frac{1}{k}}$, 则 $t = \frac{u^k - a}{b}$ $dt = \frac{k}{b} u^{k-1} du$

$\therefore \int t^q (a+bt)^{\frac{j}{k}} dt = \int \left(\frac{u^k - a}{b}\right)^q u^j \cdot \frac{k}{b} u^{k-1} du$ 化为有理函数积分

3. $p+q$ 为整数. $p = \frac{j}{k}$. 则 $\int t^q (a+bt)^p dt = \int t^{p+q} \left(\frac{a+bt}{t}\right)^{\frac{j}{k}} dt$
 此时令 $u = \left(\frac{a+bt}{t}\right)^{\frac{1}{k}}$ 即可得到有理函数

例. $I = \int \sqrt{x^2 + \frac{1}{x^2}} dx$

$I = \int x^{-1} (x^4 + 1)^{\frac{1}{2}} dx$ 令 $t = x^4$ $\frac{dt}{dx} = 4x^3 \Rightarrow dx = \frac{dt}{4x^3} = \frac{1}{4} t^{-\frac{3}{4}} dt$

$\therefore I = \int x^{-1} (t+1)^{\frac{1}{2}} \frac{1}{4} t^{-\frac{3}{4}} dt = \frac{1}{4} \int t^{-1} (t+1)^{\frac{1}{2}} dt$ 令 $u = (t+1)^{\frac{1}{2}}$ 则 $t = u^2 - 1$ $\frac{dt}{du} = 2u$

$\therefore I = \frac{1}{4} \int u \cdot (u^2 - 1)^{-1} \cdot 2u \cdot du = \frac{1}{2} \int \frac{u^2}{u^2 - 1} du = \frac{1}{2} \int \left(1 + \frac{1}{u^2 - 1}\right) du = \frac{1}{2} \int 1 du + \frac{1}{2} \int \frac{1}{u^2 - 1} du$
 $= \frac{1}{2} u + \frac{1}{4} \ln \left| \frac{u-1}{u+1} \right| + C$ 代 $u = (t+1)^{\frac{1}{2}} = (1+x^4)^{\frac{1}{2}}$ 即可